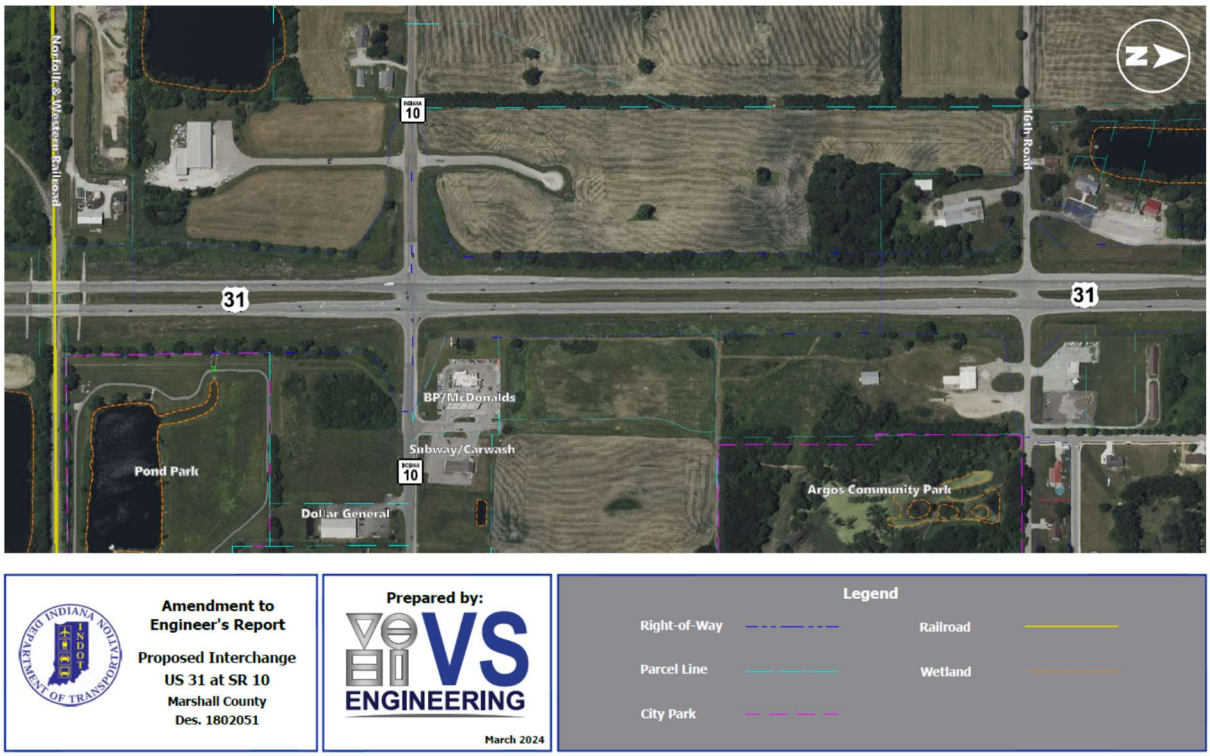
Purdue CE 498 Senior Design Course Syllabus
Spring Semester 2026
(Revised ….3/31/2026
In consort with VS Engineering
And
Indiana Department of Transportation

PROJECT TITLE: New Interchange IN US 31 at State Road 10 in Marshall County



Project Background:

The intersection of **US 31 & State Road 10 (SR 10)** near Argos, Indiana, is known as a highly dangerous, crash-prone location with numerous fatal accidents, leading the Indiana Department of Transportation (INDOT) to propose significant safety upgrades, including potential J-turns and new interchanges, as part of a major corridor project aimed at reducing severe crashes by changing traffic flow and access points for this high-speed highway corridor.

This semester, the class will function as a simulated engineering consulting firm tasked with addressing a critical infrastructure deficiency at the intersection of US 31 and State Road 10 in Marshall County, Indiana. The project focuses on upgrading a high-speed, at-grade intersection that has significant safety challenges, including rising crash severity rates and projected capacity failures. Students will navigate the full lifecycle of the design process—from an initial Feasibility Study that includes alternatives analysis, to a complex Final Design that may be subject to "Client Scope Changes" along the way. This iterative approach compels teams to evaluate real-world trade-offs between a Right-In/Right-Out configuration versus complex grade-separated interchanges, mirroring the dynamic nature of professional practice.

The project is selected specifically for its depth across all civil engineering sub-disciplines, requiring distinct contributions from every concentration. Transportation engineers will model high-speed weaving sections and conflict points; Structural engineers must design grade-separation solutions constrained by existing Norfolk & Western Railroad bridges; Geotechnical engineers will address massive earthwork variances and foundation settlement for median piers; Environmental/Hydraulics specialists will navigate Section 6(f) impacts to the Argos Community Park and design detention systems for wetland mitigation; and Construction managers will develop phasing plans to maintain US 31 traffic during complex road work. No single discipline can solve this problem in isolation; success depends on the integration of these technical systems.

This "Design Firm Simulation" is engineered to directly apply ABET Student Outcomes #2, #4, #5, and #7. By working in multidisciplinary teams, students will practice effective leadership and collaboration (Outcome #5). A project “addendum”—which alters relevant data—compels students to acquire and apply new knowledge (Outcome #7) to pivot their design strategy. Furthermore, students must apply engineering design principles (Outcome #2) not just to calculate metrics, but to make informed ethical judgments (Outcome #4) that balance economic constraints against public safety and environmental preservation, specifically regarding the sensitive community resources bordering the site.

ABET Student Outcomes:

This course is structured to facilitate the attainment of the following Accreditation Board of Engineers and Technologists (ABET) Engineering Accreditation Commission Student (Learning) Outcomes through the execution of the US 31/SR 10 Interchange Modification project:

- **Outcome #2 (Engineering Design):** Students will apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare. This is demonstrated through the iterative design (alternatives) of the interchange, where teams must calculate relevant data points, e.g., conflict points and crash reduction factors (safety), while navigating the physical constraints of the Norfolk & Western Railroad bridges and local wetland systems (welfare/environmental factors).
- **Outcome #4 (Ethics & Professional Responsibility):** Students will recognize ethical and professional responsibilities in engineering situations and make informed judgments considering global, economic, environmental, and societal contexts. This outcome is assessed through the comparative (alternative) analysis of the Phase I portion of the course where teams must weigh the economic benefits of lower-cost alternatives against the ethical necessities, i.e., preserving the Section 6(f) Argos Community Park and minimizing residential displacements, to name two relevant ethical requirements.
- **Outcome #5 (Teamwork):** Students will function effectively within a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives. By operating as a "simulated design firm," students are required to integrate distinct technical contributions—such as structural bridge design, geotechnical earthwork analysis, and construction phasing—into a single cohesive deliverable, necessitating effective cross-discipline collaboration and project management.
- **Outcome #7 (Acquisition of New Knowledge):** Students will develop an ability to acquire and apply new knowledge as needed, using appropriate learning strategies. This is explicitly exercised *between* Phase I and Phase II, where the instructors may release additional information that may alter chosen alternatives. Teams must independently digest new data and revised economic parameters, applying this new information to potentially pivot their design strategy from the originally perceived optimal solution to their final recommended alternative

ABET Design Definition:

The ABET Engineering Accreditation Commission (EAC) definition of design indicates that the process must be an iterative, decision-making exercise in which the basic sciences, mathematics and engineering sciences ae applied to convert resources into solutions. Alternative solutions must also be considered such as a “do nothing” alternative as well as other alternatives to a solution. These tenants of the ABET design definition must be an integral part of your process.

Engineering Teams:

The class will be divided into 15 engineering design teams with 10 to 11 members on a team. Each team will select a “project manager” and an “assistant project manager.” The 15 project managers and 15 assistant project managers will meet on various Tuesdays during the semester in HAMP G212 from 9:30am to 10:00 am with Professor Jacko and the Teaching Assistants (Phillip Shoopman, P.E. and Juan Nicolas Villamizar, P.E.), to discuss team progress, human resources, and other appropriate management topics. Please identify your project managers by Tuesday, January 20, 2026. Each team will be issued a time sheet outline where you will be able to record your time and activity on the project for invoicing purposes. Those time (activity) sheets will be populated and turned in weekly throughout the semester.

Convey your team’s weekly activity statements and time spent to the nearest 0.5 hr. to the TA’s every Tuesday by 11:00am. Each team will also create a company name and appropriate logo to be used on all correspondence and will convey this information to TAS Juan and Phillip by Tuesday, January 27, 2026.

Team Challenges:

In this course, your "firm" will face a series of conflicting constraints that cannot be solved by a single discipline. You are tasked with designing a high-speed interchange in a physically restricted corridor. To succeed, your team must address the following phase-specific challenges *(not considered all-inclusive; you may want to add additional, relevant content).*

PHASE I: The Feasibility Study (Weeks 1–8)
Your goal is to digest project information, evaluate at least five (5) engineering alternatives, and recommend the most viable solution based on the relevant data.

☑ Defining the Problem

- **Safety vs. Mobility:** Analyze the crash history. Does the data justify a massive infrastructure investment? Determine if the core problem is capacity (mobility) or severity (safety) based on the given metrics.
- **The "Geometric Squeeze":** Identify the hard constraints. How do you fit high-speed ramps into a footprint bounded by the Norfolk & Western Railroad bridges to the south and the Argos Community Park to the southeast?

☑ Evaluating Alternatives

- **The Matrix Analysis:** Define (design) and evaluate five (5) alternatives, of which one (1) alternative must include “no action”. Your matrix should consider topics such as costs, benefits, safety, etc., for each alternative.
- **Section 6(f) Avoidance:** Identify the legal boundaries of the Argos Community Park. Any alternative that impacts this resource requires a Section 4(f)/6(f) evaluation—does the benefit justify the environmental cost?

☑ The Recommendation (Project Management)

- **Cost-Benefit Defense:** Your firm must select a "Preferred Alternative" to present to the client. Be prepared to defend your choice: Are you recommending the cheapest option, or the one with the highest safety return?

PHASE II: Detailed Design & Adaptation (Weeks 9–16)

Following the Phase I presentations, a “data update” may be issued containing updated site data and directives. Your firm must demonstrate agility by adjusting to meet these new requirements.

☑ Possible "Scope Pivot" (All Disciplines)

- **Analyzing New Constraints:** If a data update is issued, you must assess how the new data (updated costs, fresh crash statistics, or stakeholder requests) impacts your Phase I recommendation.
- **The "Pivot" Defense:** If the update forces a change in design strategy, you must explicitly describe *why* the previous recommendation is no longer valid and how the firm will adapt.

☑ Detailed Engineering

- **Structural Rigor:** For your team’s final selected alternative, provide a detailed design of your recommendation. Make sure the design accommodates any geotechnical settlement and collision protection, if applicable.
- **The Railroad Constraint:** Verify that the final design geometry strictly avoids the Norfolk & Western Railroad right-of-way.
- **Hydraulics & Permitting:** Transition the site drainage from rural ditches to a semi-urban system (curb/gutter/sewer). Quantify wetland impacts and propose specific mitigation strategies (e.g., detention basins).

☑ Constructability & Phasing (Construction)

- **Maintenance of Traffic (MOT):** Develop a phased plan to construct a new interchange. How will you construct the design without shutting down US 31 or SR 10, or both?

- **Earthwork & Logistics:** Address the import/export balance of soil. Identify where laydown yards can be located without creating dangerous construction vehicle conflict points.

Other General Considerations

- Incorporate the AADT for the roadway.
- Consider future effects.
- Delineate the construction zone staging area.
- Estimate labor and material costs including an appropriate contingency percentage
- Manage the construction activities using a GANTT CHART highlighting the lead items
- Delineate all necessary permits identifying the max lead time permit.
- Reference appropriate Codes.

Deliverables Phase I:

The conceptual student designs will be delivered in two (2) Phases. Phase I will be devoted to project familiarization, preliminary considerations and preliminary designs executed in a documented, iterative decision-making process to include environmental, structural, geotechnical, geomatic, transportation, architectural, material specifications and hydraulic/hydrology issues. Alternative designs including a" do nothing" scenario must be shown and documented and an engineering argument made for the final selection. Preliminary cost analyses must also be included. This phase will culminate in a ***25-minute formal engineering presentation and a 25-page formal engineering report***. Each author will be clearly identified in each presentation slide as well as written section of the report. Reports to be printed single-sided and double-spaced. Only the Executive Summary and Table of Contents can be single-spaced. Appendices to be bound within the report. Three (3) hard copies of the report must be delivered on its due date (noted in subsequent calendar). YOUR TEAM NUMBER MUST BE PROMINENTLY DISPLAYED ON THE REPORT FRONT COVER, UPPER RIGHT-HAND CORNER, in 48-point font.

Deliverables Phase II:

The final conceptual student designs will again relate directly to those topics in Phase I but will include far greater detail/calculations/dimensions than in Phase I. Both a ***25-minute formal presentation and 25-page report*** will also be delivered. Each author will be clearly identified in each presentation slide as well as written section of the report. Reports to be printed single-sided

and double-spaced. Only the Executive Summary and Table of Contents can be single-spaced. Appendices to be bound within the report. Reports to be printed single-sided. Three (3) hard copies of the report must be delivered on its due date (noted in subsequent calendar). YOUR TEAM NUMBER MUST BE PROMINENTLY DISPLAYED ON THE REPORT FRONT COVER, UPPER RIGHT-HAND CORNER, in 48-point font.

Throughout the semester, each team will *submit weekly team task sheets* and *chapter summaries* from CE design textbook by Choi.

Required Text: Principles of Applied Civil Engineering Design, 2nd edition by Choi,

Ying-Kit, ASCE Press, 2017 stock number=40712, ISBN 0-7844-0712-6. This book has 25 chapters. The book is free through the ASCE Journal and can be accessed in pdf format through the Purdue library database (https://www.lib.purdue.edu/).

Suggested reference text: The Civil Engineering Handbook, Chen & Liew, CRC Press

Weekly reading /writing assignment

Each week a one-paragraph written summary for each chapter must be submitted in hard copy for the following chapters in the Choi textbook. The summary must include a reflection on what you found most personally interesting. All summaries must be held to one page and are due on the dates listed below:

Due Date	Chapters
01/20/26	1,2
01/27/26	3,4
02/03/26	5,6
02/10/26	7,8
02/17/26	9,10
02/24/26	12,13 (skip chapter 11)
03/03/26	14, 15
03/10/26	16, 17
03/16-21/26	Spring Break Week
03/24/26	18,19
03/31/26	20, 21
04/07/26	22, 23
04/14/26	24, 25

Purdue Writing Lab tutors available
Last Updated August 17, 2017, 2:40pm

Schedule appointments at http://cla.purdue.edu/wlschedule.

CE 498 Personnel:

VS Engineering, Inc

Nick Jahn, P.E., Vice President of Engineering Operations nrjahn@vsengineering.com
Tim Hollandsworth, Project Manager, VS Engr. THollandsworth@vsengineering.com
Chris Waidner, P.E. VS Engr. project manager on this intersection cwaidner@vsengineering.com

Purdue University, Lyles School of Civil Engineering:
Prof. Robert Jacko, Ph.D., P.E course director jacko@purdue.edu

Teaching Assistants:

Phillip Shoopman, P.E., Ph.D. Candidate, Environmental pshoopma@purdue.edu
Juan Nicolas Villamizar, P.E., Ph.D. Candidate, Structures jvillam@purdue.edu

And the entire CE faculty is available for consultation.

Assigned Design Rooms and Expected Hours of occupancy:

Entire class:	RHPH 172, 8:30 a.m. to 9:20 a.m.	Tuesday and Thursday
Teams 1,2,3	HAMP 1113, 9:30 to 11:20 a.m.	Tuesday and Thursday
Teams 4,5,6	HAMP 2108, 9:30 to 11:20 a.m.	Tuesday and Thursday
Teams 7,8,9	HAMP 2113, 9:30 to 11:20 a.m.	Tuesday and Thursday
Teams 10, 11,12	HAMP 2123, 9:30 to 11:20 a.m.	Tuesday and Thursday
Teams 13, 14,15	HAMP 3153, 9:30 to 11:20 a.m.	Tuesday and Thursday

Note: (a) These same office hours also apply to the Olson Design Lab in room 3154 for all teams. (b) Each team is required to meet in the above-mentioned design rooms from at least 9:30am-10:00am on Tuesdays and Thursdays when the instructors make the initial rounds.

Grading:

Chapter summaries: 5%

Presentation procedures and grading: For Phase I, each team will select five (5) members to make the oral presentation to the "owners and observers" (faculty and students). The non-speakers will also position themselves at the front of the Beck Center conference room adjacent to the five (5) speakers. The five (5) speakers will have a maximum of twenty-five (25) minutes for their presentations and immediately following the presentations there will be a ten (10) minute Q & A session for owners and observers. Questions must be directed to a speaker who then must redirect the question to one of the non-speaker team members and the speaker can use discretion during this non-speaker answer period. The final score for that speaker is affected by the answer to the question. An overall team score for their presentation will be determined by considering each individual speaker's score that is averaged with an overall team score. That result will then be assigned to all team members. Phase II presentations will be assessed similarly, however, the same speakers for phase I cannot be speakers for phase II.

Written Report Procedures and Grading Phase I and II: Reports are limited to twenty-five (25) pages, double-spaced. Only the Executive Summary and Table of Contents can be single-spaced. Authors of sections in the report must be identified with their last names shown in the title section. Appendices are to be bound within the report. Three (3) hard copies of the report must be delivered on its due date (noted in subsequent calendar). YOUR TEAM NUMBER MUST BE PROMINENTLY DISPLAYED ON THE FRONT COVER OF THE REPORT in 48-point font in the upper right-hand corner.

Overall Phase I Grade: Preliminary (written) Engineering Report 20%, oral presentation 15%.

Overall Phase II Grade: Final (written) Design Report 40%, oral presentation 20%.

Final Grading: Scoring of each deliverable will be based on a 100 point scale.

The overall course grade will be the weighted average of chapter summaries (5%), Phase I (35%) and Phase II (60%). Note that the CCE school policy states that a C- final grade average, (e.g. <74.0 points) requires a course re-take.

Attendance and I-Phones: Attendance by all is required at the first lecture hour on Tuesday and Thursdays when the **use of I-Phones and computers is prohibited** during the lecture hour

except for notetaking, as it is highly disrespectful to the guest speakers. Also, there is no phone and computer usage during the 3 hour-long formal engineering presentation sessions culminating in Phase I and II. For the remaining two (2) hours of design lab each Tuesday and Thursday, you are expected to be working with your team; a proposed progress schedule is included with this syllabus. **The penalty for an unexcused absence will be removal of one point from your final course average grade for each unexcused absence. If you would like to be excused for a legitimate reason (illness, job interview, conference, etc.), email Prof. Jacko and TAs Phillip Shoopman and Juan Nicolas Villamizar prior to the class you want to be excused from and state the reason.**

Check regularly the latest syllabus, class presentations, and other class related information.

***OneDrive:* Additional information on our project will be posted on a shared folder.**

(***) You will receive an email with a link on how to access the shared folder (use your Office 365 account).

Begin reading about the Project, its history, existing technical condition and become familiar with the data regarding the site. Visit the following link below:

CCE498 Resources

FIRST HOUR CALENDAR (8:30-9:30am)

Note: The following calendar dates and times are subject to change. You will be notified in advance of these changes.

Phase I: Preliminary Engineering Report (note: all the following classes begin at 8:30am in RHPH 172 and then proceed to your assigned HAMP lab rooms for the remaining two. Oral engineering team presentations all begin at 8:00 am**.**

Date	Activity
01/13/26 Tu	**8:30am classroom RHPH 172** Personnel introductions; Project Introduction: A New Interchange at IN US 31at State Road 10 in Marshall County. Chris Waidner, P.E. Indiana Transportation Director and project manager on this US 31 & SR 10 intersection, VS

Engineering to expand on project. Student Team assignments are provided, as well as University Emergency Safety Instructions discussed Confirmed 12/16/2026

01/15/26 Th 8:30 to 9:20am, Taylor Ruble, P.E., RSP, Manager Office of Traffic Safety, Traffic Engineering Department, Indiana Department of Transportation will discuss the intersection. Confirmed 12/15/2025

01/20/26 Tu Professor Brandon Fulk to explain CCE career placement opportunities. Confirmed 12/15/2026

Project Managers and Assistant Project Managers meeting G212 9:30 to 10:00am

01/22/26 Th Professor Andrew Tarko of Purdue's Transportation Engineering Group to discuss intersection safety. Confirmed 12/17/2025

01/27/26 Tu Professor Venkatesh Merwade of our Hydraulics Group to discuss water sheds, rainfall as applied to our project. Confirmed 12/18/2025

01/29/26 Th Professor Dennis Lyn from the Purdue Hydraulics Specialty Area to discuss roadway intersection drainage, overtopping and other hydraulic aspects of our project. pending

Project Managers and Assistant Project Managers meeting G212 9:30 to 10:00am

02/03/26Tu Phillip Shoopman, P.E., Ph.D. candidate and our TA to discuss environmental aspects of our project. Following lecture, students released to attend the Career Fair.

02/05/26 Th No guest speaker this morning at 8:30am so report to your design rooms at 9:30am.

02/10/26Tu Juan Villamizar, P.E., PhD Candidate from Purdue's Structural Engineering Group and our TA to discuss structural aspects involved in our project.

02/12/26 Th Professor Jacko to discuss the finer points on giving power point presentations, report writing, and binding/organization requirements. Team design summary statements due today: **250** words describing your team's management strategy.

02/17/26 Tu Ron Klemencic, P.E., S.E., Hon. AIA, NAE, Chairman & CEO, Magnusson Klemencic, Associates, Seattle and Chicago to address our senior design teams emphasizing the importance of in-house team collaboration and innovative design Confirmed 12/16/2026

02/19/26 Th 8:30am **Class Photos:** G212 Conference room. Individual class photos will be taken so business dress is required, i.e., suits, ties, etc. Ms. Jaimee Barr will give you the details as to the exact times etc.

02/24/26 Tu **Phase I Oral Presentations will begin at 8:00 am at the Beck Agricultural Center on HWY 52 West**

Procedure: Following each team's presentation, the 10-minute Q & A session will begin. Owners and Observers will participate. Location: BECK Agricultural Center on Hwy 52 West.

02/24/26 Tu **Teams 1, 2, 3, 4, 5** Each team has 25 minutes for the presentation and 10 minutes for questions from the faculty and students. All team members must be present. All observing teams and their members are required to attend. Beck Ag Center Hwy 52 west.

02/26/26 Th **Teams 6, 7, 8, 9, 10** Each team has 25 minutes for the presentation and 10 minutes for questions from the faculty and students. All team members must be present. All observing teams and their members are required to attend. Beck Ag Center Hwy 52 west.

03/03/26 Tu **Teams 11,12,13,14,15** Each team has 25 minutes for the presentation and 10 minutes for questions from the faculty and students. All team members must be present. All observing teams and their members are required to attend. Beck Ag Center Hwy 52 west.

03/05/26 Th **Phase I Written reports due from all teams at 11:00 am in the Olson Lab. Report to be submitted in three (3) paper copies and one digital copy** (25 pages max single-sided, appendices to be bound in a separate rear section of the report). No formal meeting.

CALENDAR: Phase II Begins

Begin Phase II Tuesday, March 11, 2025

03/10/26 Tu Chris Waidner Project Manager and Joey Martin, Research Engineer, VS Engineering, Northwest Indiana office to discuss final design considerations. Confirmed 12/16/2026

03/12/26 Th Professor Antonio Bobet from our Geotechnical Engineering Group will discuss soil conditions and geotechnical design considerations at our site. Confirmed 2/11/2026.

03/16-21/26 Spring Break

03/24/26 Tu Claude Goguen, P.E., LEED AP, Director of Sustainability and Technical Education National Precast Concrete Assn., Indianapolis and Mr. Steve Smart, Technical Resource Engineer, County Materials to discuss the use of precast drainage and structural elements in the new intersection. Confirmed 3/10/2026

03/26/26 Th No guest lecture hour at 8:30am so please report in your design rooms at 9:30am and you will receive your graded phase 1 reports back along with feedback. Then during rounds we will answer questions.

03/31/26 Tu Postdoc Researcher on pavements from Professor John Haddock's staff, Dr. Mohammad Ali Notani will discuss pavement design, i.e., asphalt or concrete. Confirmed 2/3/2026

Project Managers and Assistant Project Managers meeting G212 9:30 to 10:00am

04/02/26 Th CEAAA Day 8:30am, Civil Engineering Alumni Achievement Awards. All students meet as normal in RHPH 172 to meet and find out how our honored alums became so successful.

04/07/26 Tu Professor Ted Weidner of the Purdue Construction Engineering Management Group to discuss MOT during road construction among other relevant items to our project. Confirmed 3/31/2026

04/09/26 Th Alum guest speaker Emily Whelchel, Project Engineer with ARSEE Engineers of Indianapolis to discuss the benefits of Professional Engineering Licensure.

04/14/26 Tu Special guest speaker Jason Williamson is a Purdue Civil and Construction Engineering alumnus selected for the all-Boilermaker "Purdue 1" suborbital spaceflight, scheduled for 2027. He currently serves as a senior vice president at Dunaway. BSCE 1997. Confirmed 3/20/2026

Project Managers and Assistant Project Managers meeting G212 9:30 to 10:00am

04/16/26 Th No formal guest lecture hour. Work in teams to put the finishing touches on your Phase II presentations. Prof. Jacko, Juan Villamizar and Phillip Shoopman will do rounds in HAMP beginning at 9:30am to answer questions and provide advice.

04/21/26 Tu **Phase II Engineering Presentations Begin, 8:00am at the Beck Agricultural Center on HWY 52 West**

Procedure: Following each team's presentation, the 10-minute Q & A session will begin. Owners and Observers will participate. Location: BECK Agricultural Center on Hwy 52 West.

04/21/26 Tu **Teams 11,12,13,14,15. Each** team has 25 minutes for the presentation and 10 minutes for questions from the faculty. All observing teams and their members are required to attend.

04/23/26 Th **Teams 6,7,8,9,10.** Each team has 25 minutes for the presentation and 10 minutes for questions from the faculty and students. Location: All observing teams and their members are required to attend.

04/28/26 Tu T**eams 1,2,3,4,5.** Each team has 25 minutes for the presentation and 10 minutes for questions from the faculty and students. Location: All observing teams and their members are required to attend.

04/30/26 Th **11:00am. Final class meeting** in the **Olson Lab, HAMP 3154.** Ms. Jaimee Barr will present information on filling out your exit survey, also-------

Written reports due from all teams. Report to be submitted in three (3) paper copies double spaced, single-sided and a digital copy of the report. (25 pages max, appendices to be bound in a separate rear section of the report). Page 26 will contain the reference list.

In the event of a major campus emergency, course requirements, deadlines and grading percentages are subject to changes that may be necessitated by a revised semester calendar or other circumstances. For more information send an email request to jacko@purdue.edu

Course materials are undergoing remediation per the ADA Title II Ruling. Efforts are being made to the best of my ability to ensure that course materials are ADA compliant; however, these materials have not been vetted by a third party. If you are having difficulties accessing materials, please contact me immediately for support. Any materials needing further remediation will be completed by the beginning of Fall 2026.